

AMBIENT SYSTEMS

Climate At Your Service

www.ambientsystems.ai

Cyber-Physical Systems (CPS) are integrations of computation, networking, and physical processes.

Swarm Intelligence

Edge Devices

Cyber-security

Robotics



Custom Applications

CPS requires communication and intelligent mechanisms to fulfil competing priorities



Indoor Farms

Parameter Schedule
Harvest Prediction



Hospitals

Zone Control Bacteria
Detection



Retail Stores

Customer Experience
Footfall and Heatmaps Analysis

Problem

There is increasing demand for intelligent machines that can interact with the environment around them

1. Limited Resources

It’s hard to find highly trained and experienced operators who can optimize multiple intelligent systems.

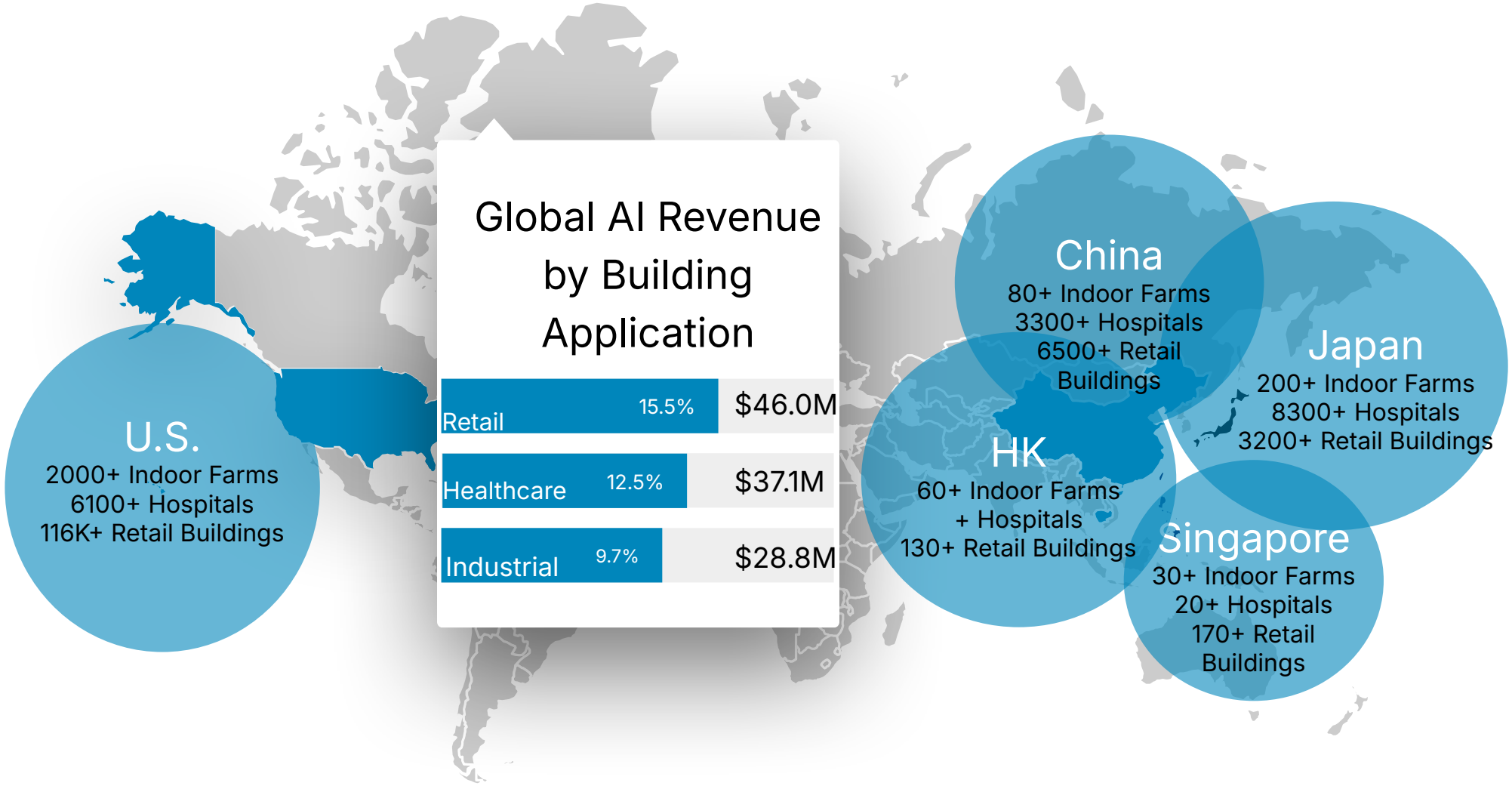
2. Technological Complexity

Many operate with outdated legacy systems and physical equipment that lack interoperability.

3. Competing Priorities

Different physical systems have different requirements based on the unique needs of occupants and operators.

Unlocking USD$110M of revenue across specialised segments



AI Platform Backed by IP





Sense

MVP sensors co-designed with manufacturing partners



Optimize

NUS National University of Singapore

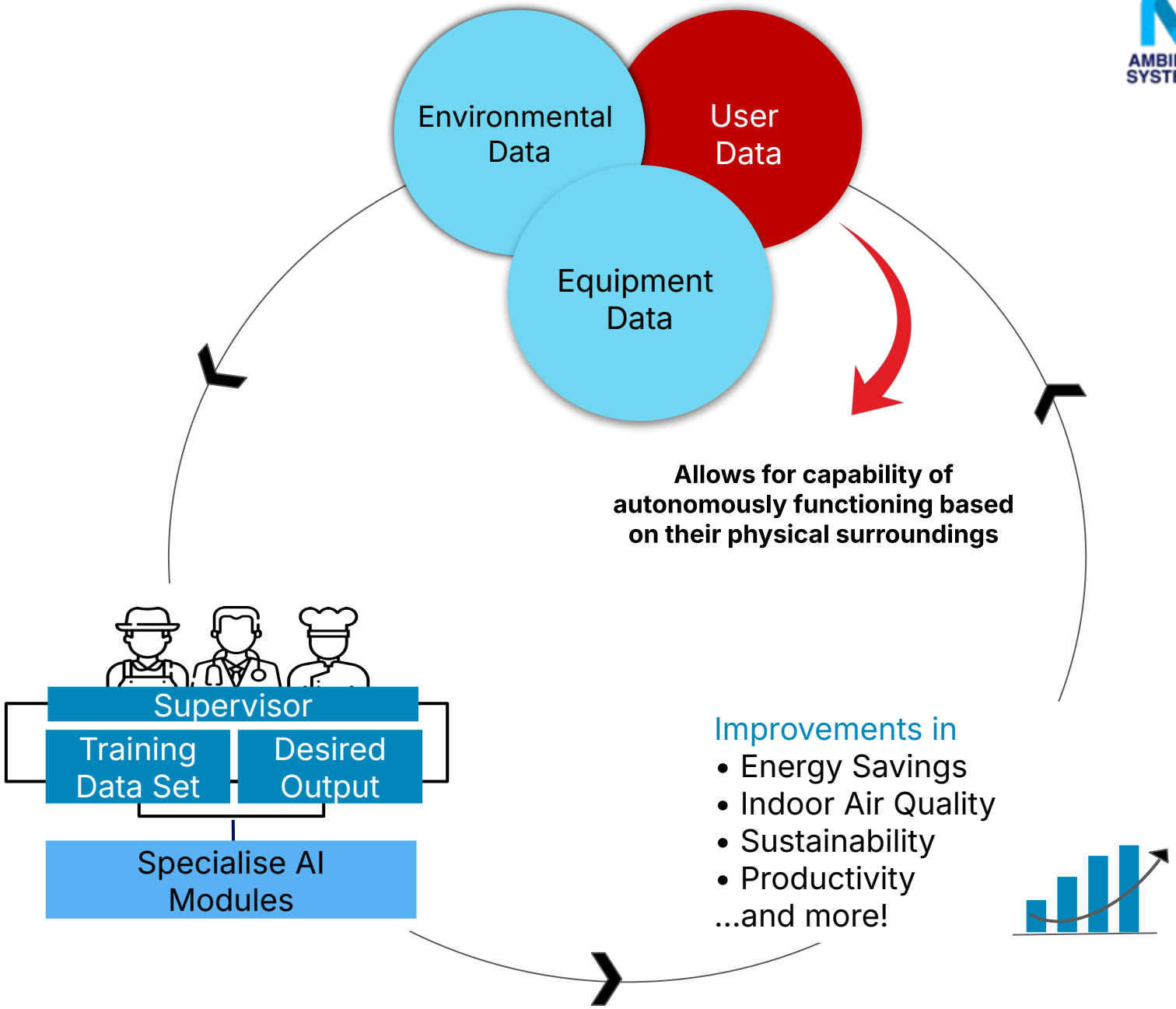
Patent Pending Invention from National University of Singapore (NUS)



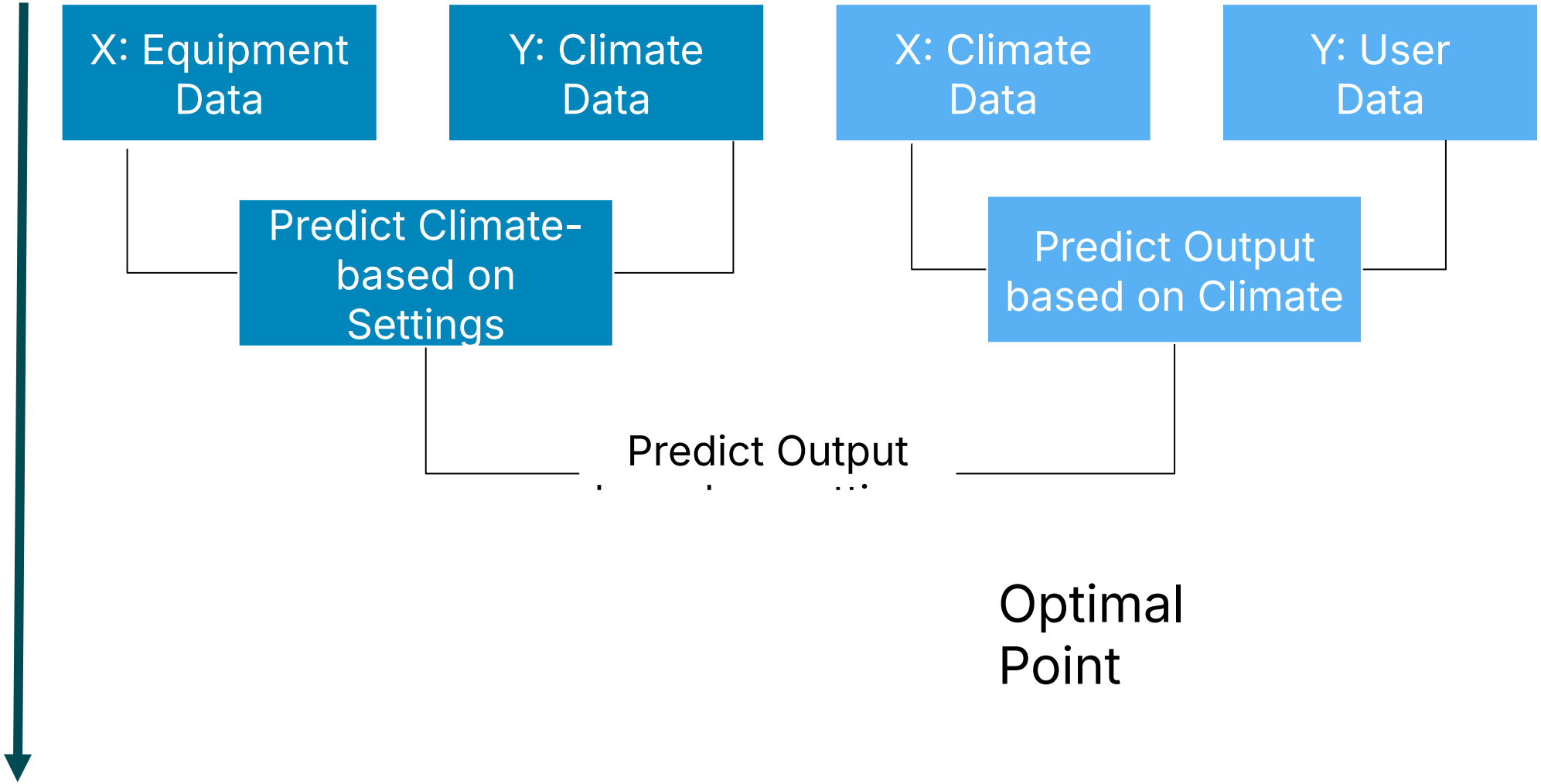
Control

Offering full-stack web application after development

System Optimization beyond Energy Savings



Underlying Magic: for Management of Ph



Singapore patent: 10202105064T

Increasing Energy Demand and Consumption



Source: Our World in Data

In the built environment, it is imperative for buildings to be even more energy efficient and drive greater reduction in

Increasing Focus on Sustainability

90%

of consumers are concerned about the Environment

Source: Sustainability Matters Now More Than Ever for Consumer Companies BCG, 2020

50%

of CXOs see tackling climate change as their generation's top priority

Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020

Ambitious targets to green the built environment in Singapore by 2030:

80% green buildings

80% improvements in energy efficiency

80% new buildings to be super lower energy

Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020

Smart and Healthy Buildings Brings Value

Smart office buildings can potentially garner a **37%**

rent premium per nett square feet, compared to buildings without smart characteristics

Source: How Much Are Smart Buildings Really Worth? NAIOP Massachusetts, 2018

Healthy buildings increase productivity by **US$6,600 per employee**

translating to cost savings and increased revenue for tenants

Source: The Best Talent will Demand the Smartest Buildings Memoori Smart Building Research, 2018

Building analytics can reliably drive **3-9%**

energy savings annually, a figure expected to persist or even increase

Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020

Ambient aims to be a turnkey solution for CPS

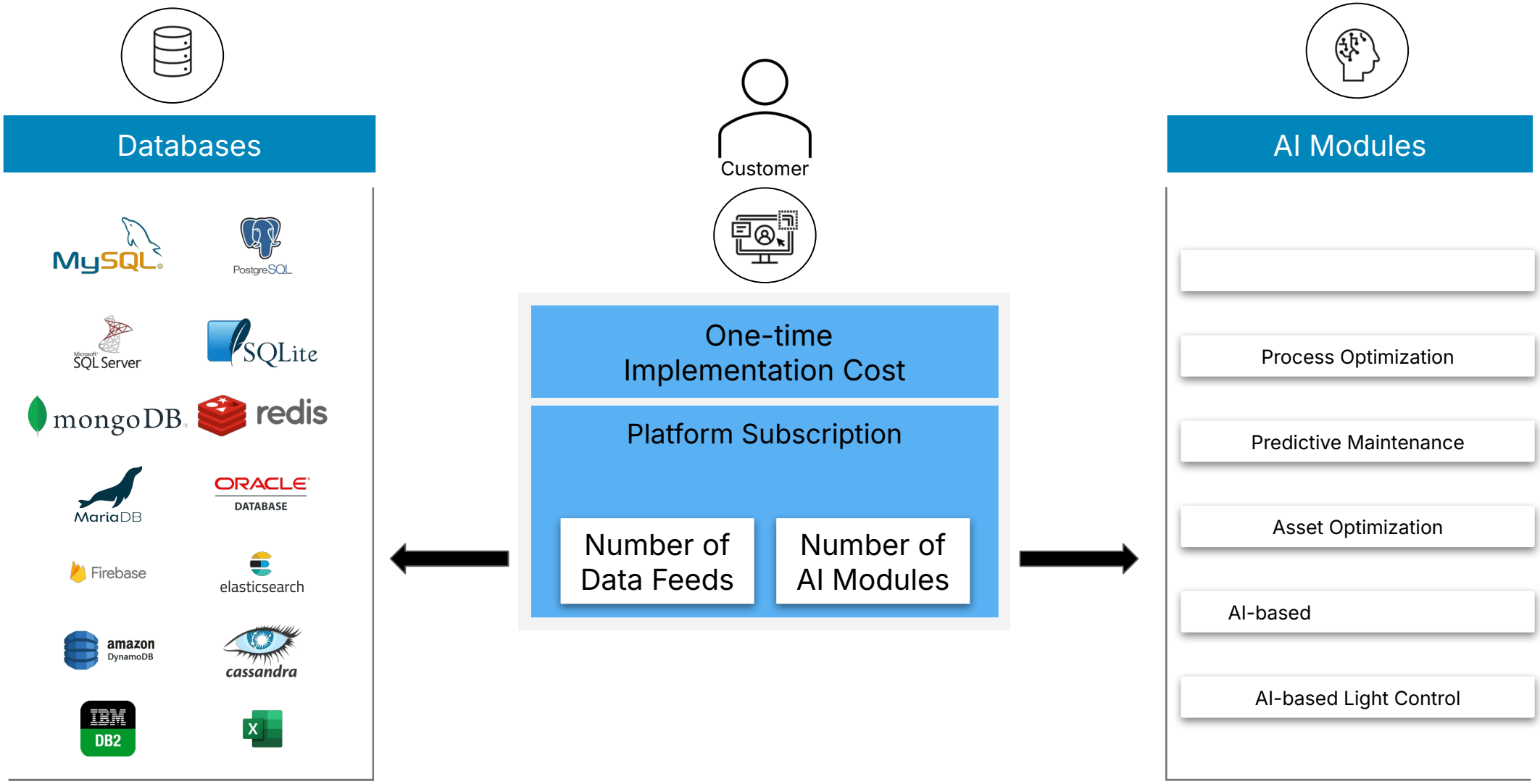
(Global)					
Name	**Ambient**	**Olive**	**Indigo Ag**	**Crowdstrike**	**Tempus**
Description	Ambient is an enterprise artificial intelligence company looking to provide applied AI to cyber-physical systems. The company software offers automation-enabled planning, scheduling, predictive maintenance, efficiency of decision-making.	Developer of process automation software designed to build meaningful artificial intelligence that empowers and scales humans. The company's software acts as a router between systems and data by automating repetitive, high-volume tasks and workflows.	Provider of plant microbiome agricultural services designed to increase crop yield. The company predict which microbes are most beneficial to the health of the crops and apply these specially selected microbes in the form of a seed coating, enabling farmers to reduce risk and increase profitability.	CrowdStrike Holdings, Inc. is a cybersecurity vendor that specializes in endpoint protection, threat intelligence and hunting, attack remediation, and offers various solutions to supplement security and network operations teams.	Developer of a healthcare data analytics platform designed to improve patient outcomes. The company empower users to make real-time and data-driven decisions, enabling physicians to deliver personalized care for patients through an interactive analytical and machine learning platform.
Segments	Enterprise AI	HealthTech	AgTech, Climate Tech	Cybersecurity	HealthTech
Valuation (USD M)	Undisclosed	3,990 (Jun 2021)	2,760 (Jun 2020)	56,670 (Oct 2021)	8,100 (Dec 2020)
Series A Round (USD M)		5 (Aug 2013)	7.5 (Nov 2014)	26.51 (Feb 2012)	10 (Sep 2015)
End-point Detections	✓			✓	
Cyber-security Middleware	✓			✓	
Robotic Process Automation	✓	✓			
Physical Systems Network	✓	✓			
Data-library Assets	✓	✓	✓		✓

Agtech and Cyber-security focus in SEA

(Global)

	Ambient	**Sero**	**Agribuddy**	**Horangi**	**Cyfirma**
Name					
Description	Ambient is an enterprise artificial intelligence company looking to provide applied AI to cyber-physical systems. The company software offers automation-enabled planning, scheduling, predictive maintenance, efficiency of decision-making.	Provider of a crop intelligence platform intended to optimize rice production with full-cycle data analytics. The company's platform enables farmers to optimize their rice production by tracking, managing and keeping their crops healthy.	Developer of an ag-tech platform designed to assist farmers in analyzing farm-related information and improve crop yields through collective intelligence. The company's enables rural farmers to manage farms and adopt new farming techniques and be assisted with asset management support services.	Provider of full-stack cybersecurity platform. The company's platform hunts for advanced persistent threats and permits an organization's software development lifecycle to be secure, enabling businesses and corporate clients to understand and respond to issues related to cybersecurity.	Developer and provider of cybersecurity technology that provides threat visibility and intelligence products and services. The company's enables clients to identify threats accurately, receive early warnings of impending cyberattacks to protect their IT infrastructure from breach.
Segments	Agtech, Cyber-security				
(USD M)	Undisclosed	0.33 (Nov 2016)	Undisclosed	Undisclosed	Undisclosed
Series A Round (USD M)	-	0.02 (Nov 2018)	3.53 (Oct 2018)	23.1 (Mar 2020)	8M (Feb 2020)
End-point Detections	✓	✓		✓	✓
Cyber-security Middleware	✓			✓	✓
Robotic Process Automation	✓				
Physical Systems Network	✓				
Proprietary Data Assets	✓	✓	✓	✓	✓

B2B SaaS Model



Usage-based Pricing Plan

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One-time Implementation Cost
Installation of sensors and equipment integration

by Project Basis

Basic
SGD$1,000

- Custom data analytics

3 Data Feeds/mo
1 AI modules/mo

Advanced
SGD$1,500

- Basic + two more data feeds
- one more

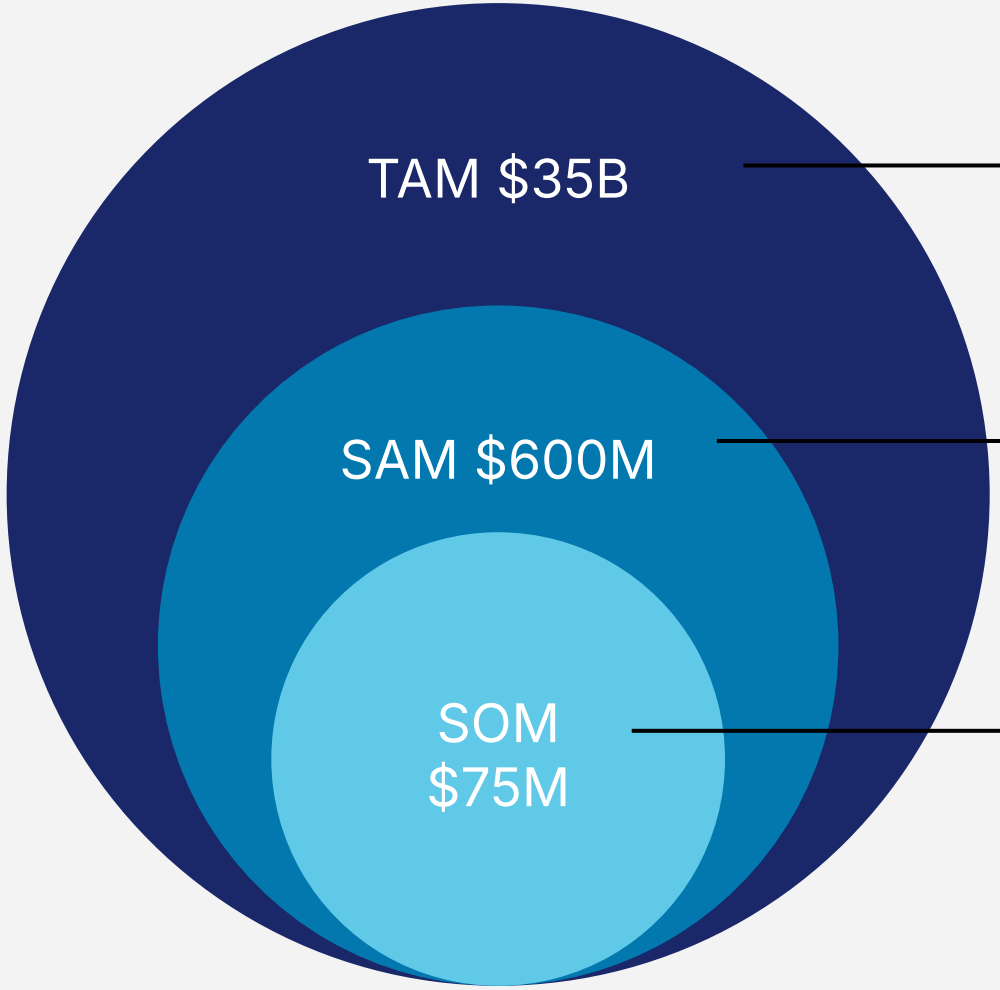
5 Data Feeds/mo
2 AI modules/mo

Premium
SGD$2,000

- Advanced + two more data feeds
- Choice of three AI modules
- Choice of three AI modules

7 Data Feeds/mo
3 AI modules/mo

Opportunity



Global AI Market
(CAGR 42%)

Global AI for Buildings & Homes Market
(CAGR 36%)

Target Market: Retail, Healthcare, and Industrial Buildings in APAC and North America

Source: Frost & Sullivan

Go-to-market Strategy

Monitor

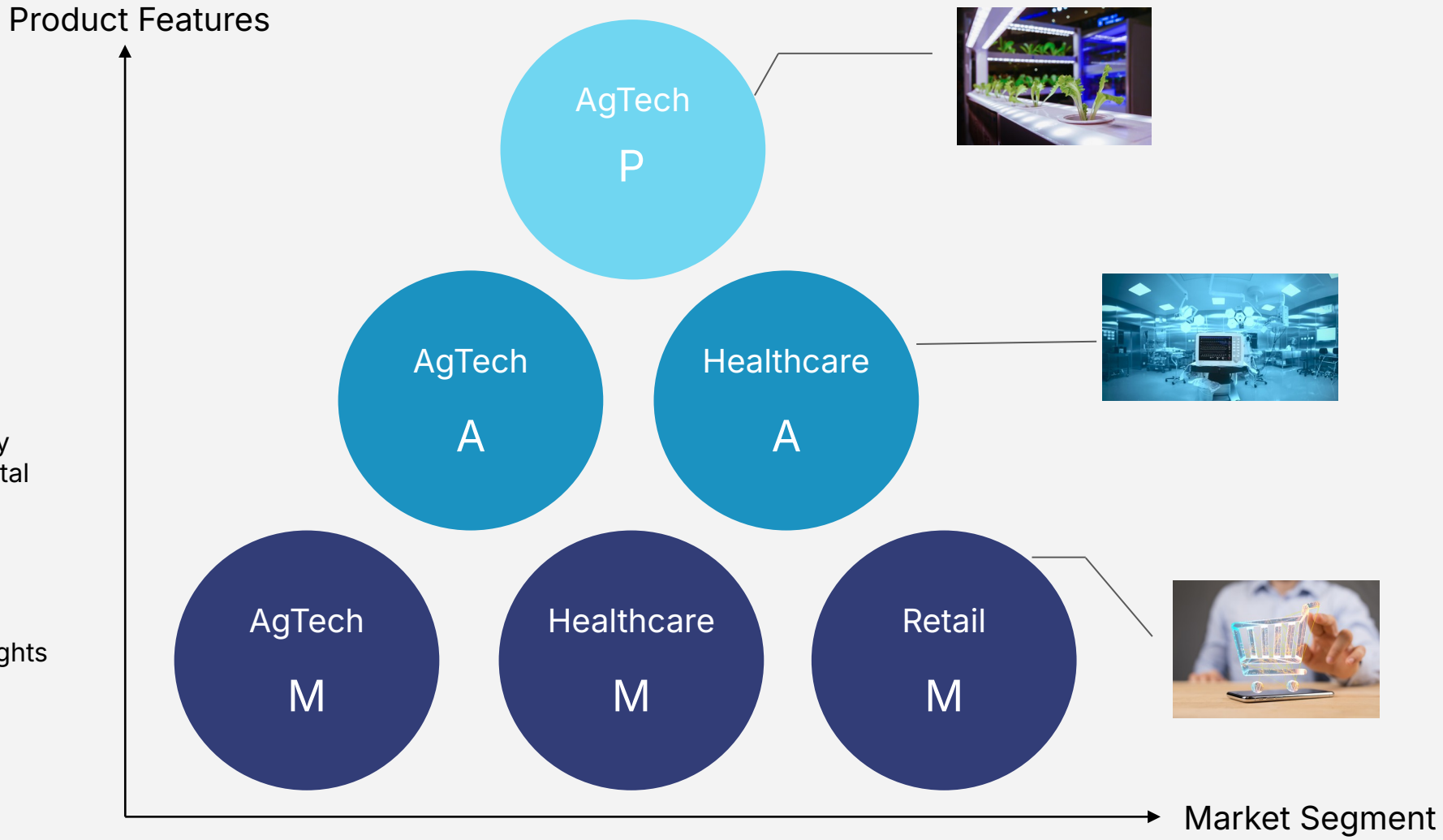
- Real-time data
- Online dashboards
- Push Alerts

Analyze

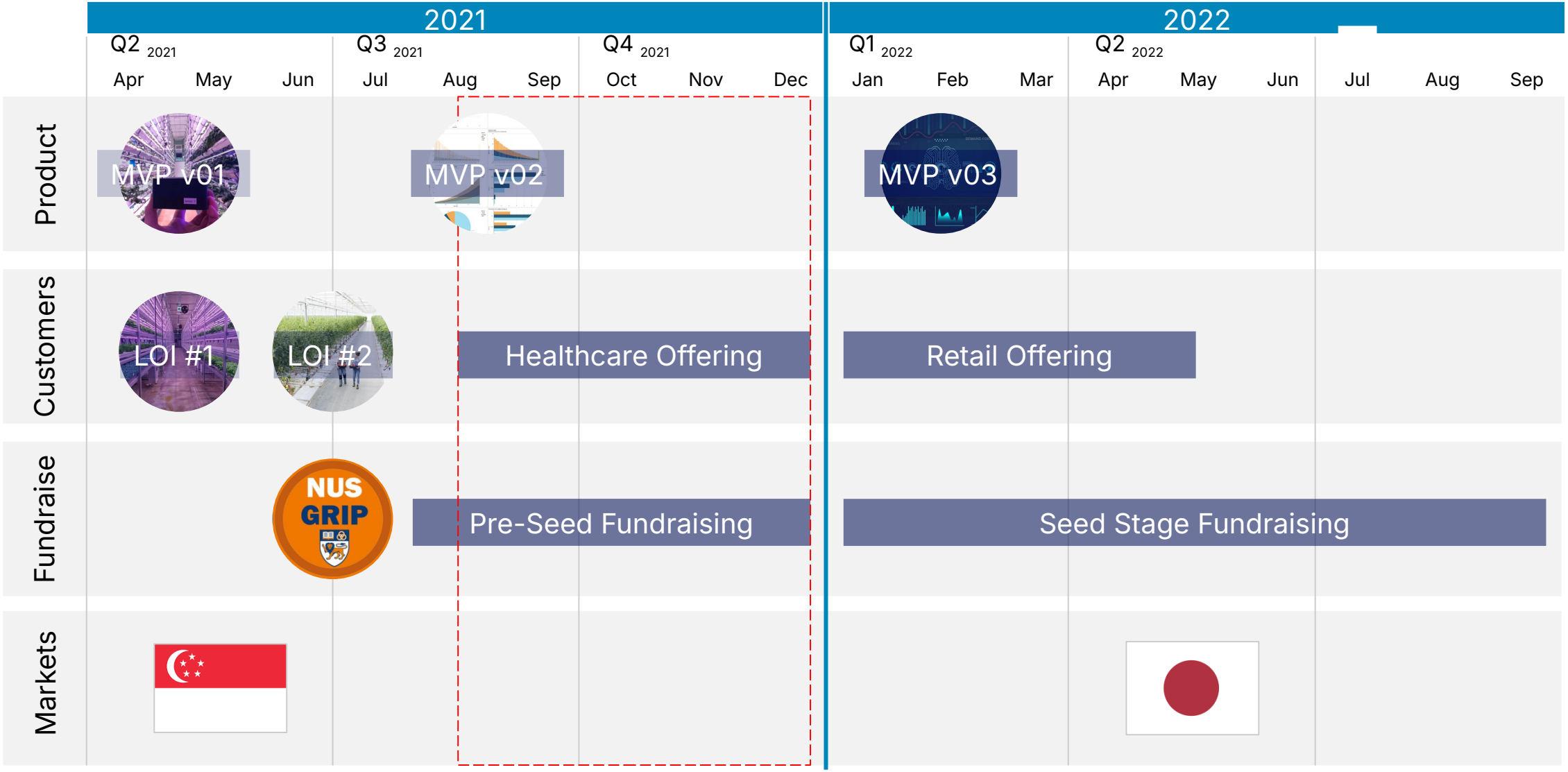
- Full HVAC connectivity
- Real-time environmental adjustments

Prescribe

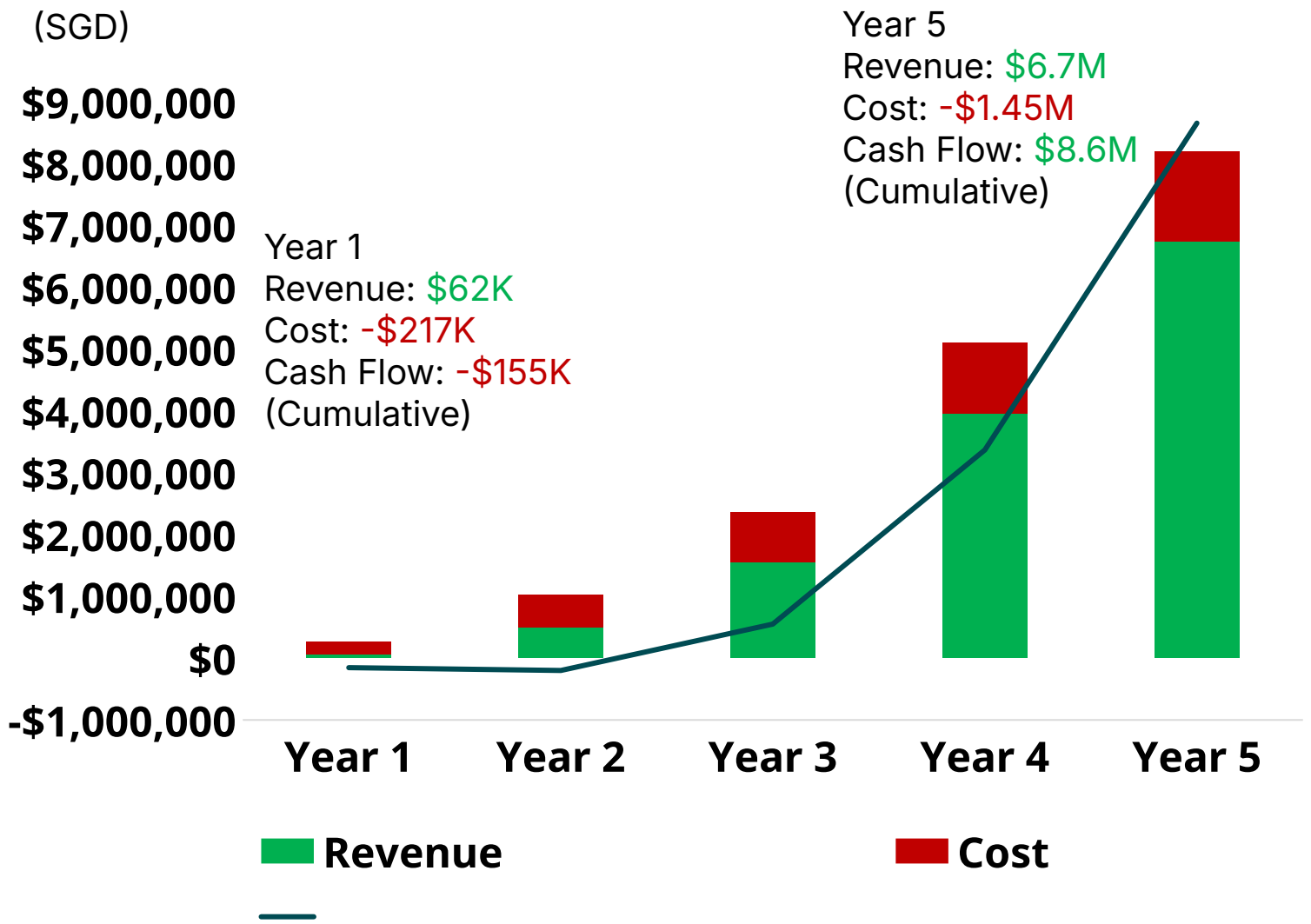
- Integrate user data
- Obtain actionable insights according to needs



Traction & Current Milestones



Revenue Forecast



81.2%
Average Gross Profit

$8.5M
NPV

Our Team

CEO
Background in Engineering and Computer Science

CSO
Experience in fundraising and strategy at APAC Unicorn startup

Data Analyst
Creative Technologist with background in visual communications

Project Management Intern
Marketing at Unilever China

Developer
Built websites and applications for top brands in APAC

Thank You

Visit us at: www.ambientsystems.ai

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Appendix

One step ahead with AI-enabled prescriptive maintenance

AMBIENT SYSTEMS
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Artificial Intelligence	Machine Learning	Deep Learning
		Data points gathered 
Automated Offerings • Building Performance Optimisation • Building Process Optimization	Smart Offerings • Asset Optimization • AI-based • AI-based Lighting Control	Cognitive Offerings • Predictive Maintenance

Identify general issues

Understand energy consumption by analysing primary datapoints

Personalised control

Understand energy flow and building occupancy

Collect external datapoints

Indoor Farming Overview

Improve crop yield in a market where only **51% of Indoor Farms are profitable**

24% reduction in loss of crops due to poor growing environment

Real-time improvements in farm operations efficiency resulting in **0.5 - 1.5 years** payback period



Indoor farm operators seeking to optimize crop performance through customized insights across key parameters.

Healthcare Overview

Prevent **65-70%** of hospital-associated infections costing up to USD11B

Hospitals with superior patient experience achieve **50%** more in net margins

Improve air filtration and recirculation on hospital premise to prevent nosocomial outbreaks of COVID-19 **(Mortality rates >60%)**



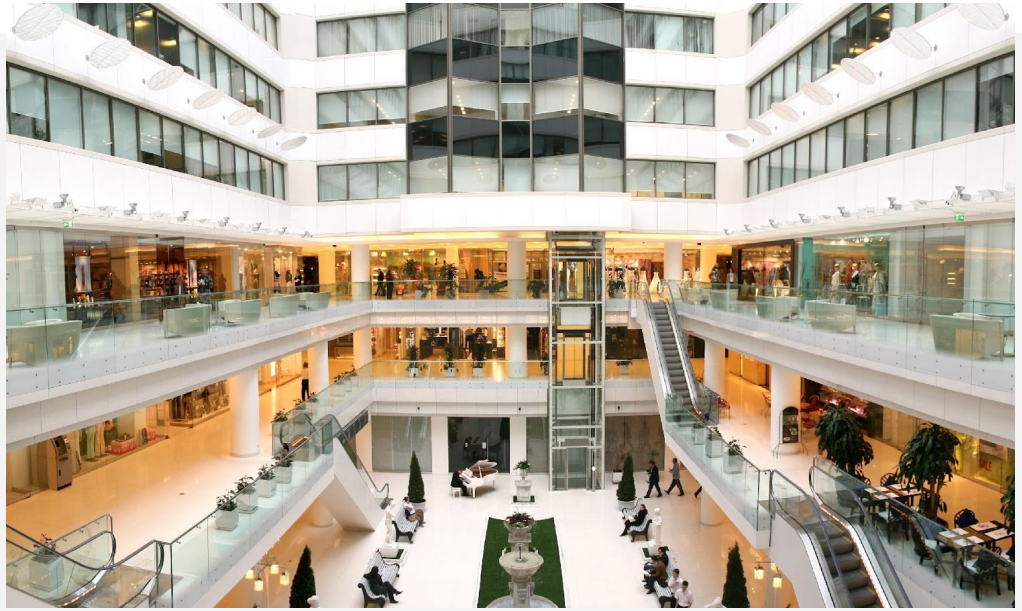
Hospitals looking to improve patient well-being hrough monitoring of indoor air quality

Retail Overview

70% of customers avoid retail stores if they are too hot, too cold or too stuffy

Using retail heatmaps to optimise retail layout resulting in **5%** increase in sales

Improve in-store conversions by more than **10%** by analysing customer footfall trends



Retailers looking to optimize customer experiences through improving retail comfort and analysing footfall behavior